TAO person action and pose classification

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1. Case objective and processing flow
2. Checks before running
3. Starting the action recognition
4. Real-time detection with a USB camera
5. Real-time detection with a CSI camera
6. Common issues
7. References

> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case objective and processing flow

This case first detects and tracks the people, then estimates the key points, and finally recognizes the action with a time series. A single frame cannot verify the complete action classification function.

text

```
Video/USB/CSI → PeopleNet → NvDCF track ID → BodyPose3DNet 34 points
             → temporal window → PoseClassificationNet (ST-GCN) → action category
```

Each target needs to accumulate about 300 frames of key points; at 30 FPS the action category may only become stable after about 10 seconds.

2. Checks before running

Video mode:

bash

```
cd ~/deepstream_tao_apps

# Confirm the model directories
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/bodypose3dnet_vdeployable_accuracy_onnx_1.0/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/poseclassificationnet_vdeployable_onnx_v1.0/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplenet_vdeployable_quantized_onnx_v2.6.3/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/reidentificationnet_vdeployable_v1.2/

# Confirm the configuration files
ls -la ~/deepstream_tao_apps/configs/app/bodypose_full.yaml
ls -la ~/deepstream_tao_apps/configs/app/bodypose_usb.yaml
ls -la ~/deepstream_tao_apps/configs/app/bodypose_csi.yaml
```

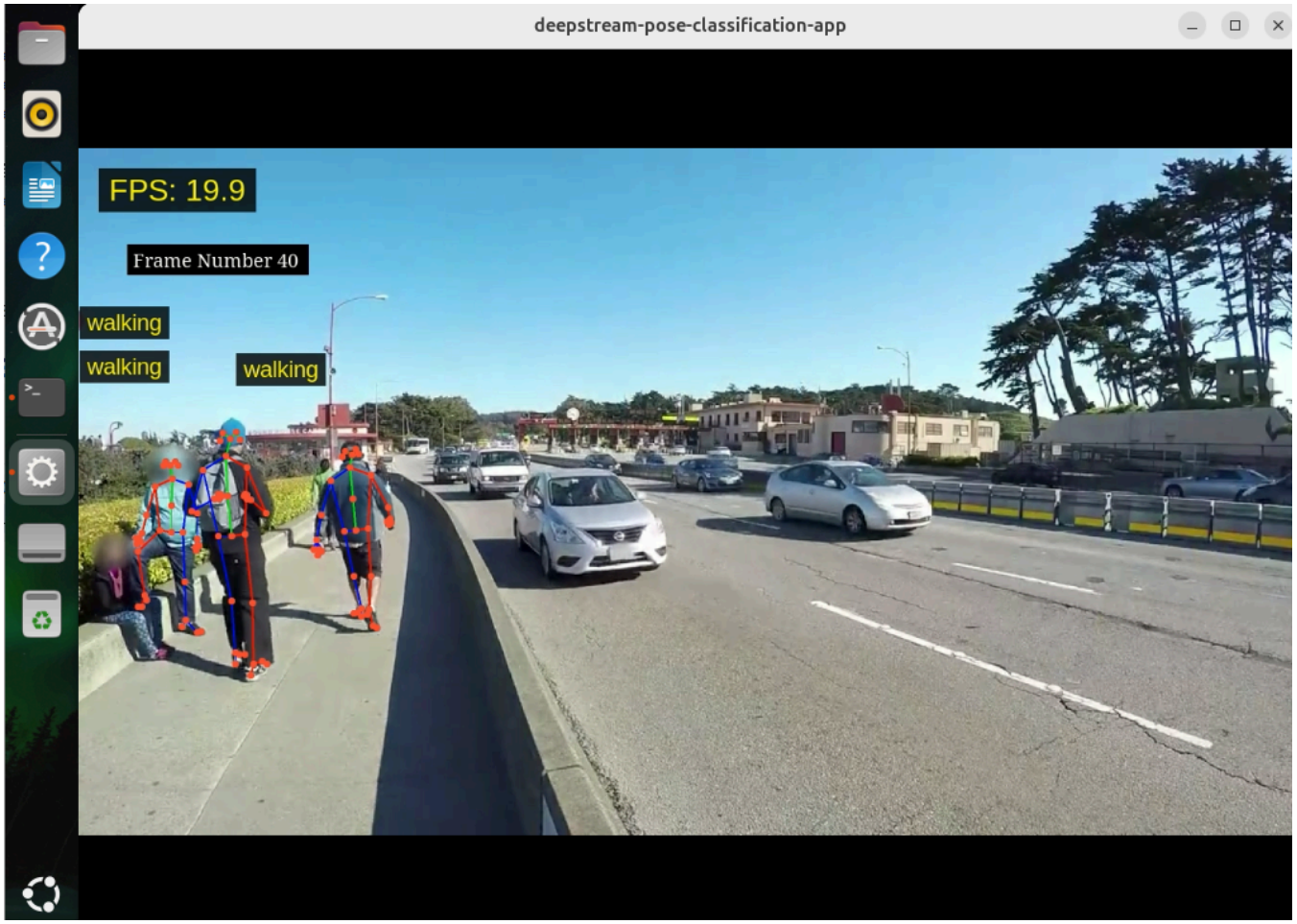
```
# Confirm the compiled artefact of the dedicated app
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream-pose-classification/deepstream-pose-classification-app
```

The camera mode has to run in a Jetson desktop/VNC session with the camera connected. A target that has just entered the view needs to wait for the temporal window to accumulate.

3. Starting the action recognition

Offline video:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_poseclass.sh
```



It uses the `~/deepstream_tao_apps/configs/app/bodypose_full.yaml` configuration, and the default input is two video streams (walk + run).

Or pass the parameter directly to replace the video:

```
~/deepstream_tao_demo/run_poseclass.sh /home/jetson/Videos/your_video.mp4
```

The person boxes, the skeleton and the action label come from the detection, the pose estimation and the temporal classification respectively. After the EOS of the input source, check the output:

```
# Location of the output MP4
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream-pose-classification/out.mp4
```

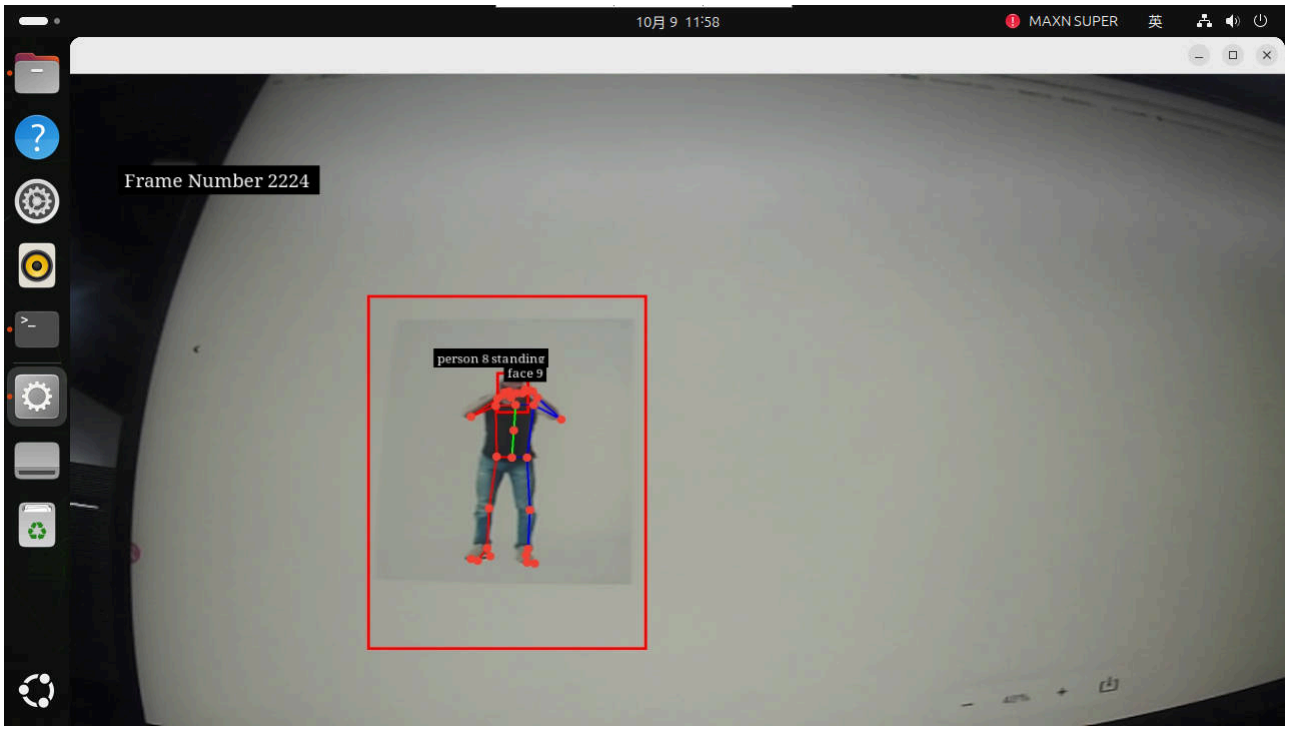
To replace the test video: edit the `source-list.list` in `~/deepstream_tao_apps/configs/app/bodypose_full.yaml`; the two videos are separated by a semicolon. Prepare material with a sufficient duration that contains continuous actions, and keep the original file. If you change the frame rate or the clip is too short, the temporal window may not be able to give a stable category.

4. Real-time detection with a USB camera

The USB entry point of the course reuses the real-time camera source:

```
v4l2-ctl -d /dev/video0 --list-formats-ext

cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_poseclass.sh usb
```



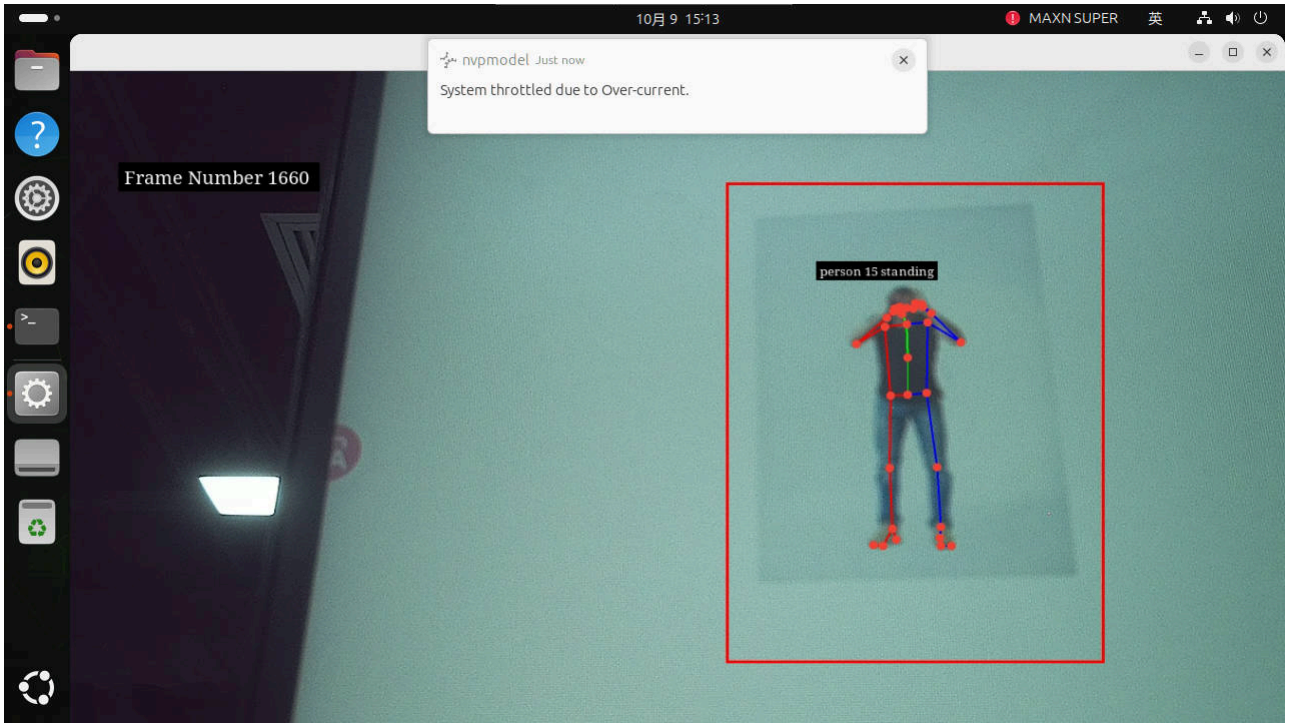
Make sure a body stays in the view and leave time for the key points to accumulate. The USB real-time mode is used for the preview and does not generate `out.mp4`. Press `Ctrl+C` once and wait for the source EOS and the release of the device.

5. Real-time detection with a CSI camera

The CSI entry point uses Argus, so you can try it after connecting a supported module and confirming the driver/the service:

Start:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_poseclass.sh csi
```



CSI supports the real-time action preview, but the key points still have to accumulate continuously. The output file of the video mode cannot directly serve as the acceptance criterion of the camera mode.

6. Common issues

Symptom	Check and handling
No action label right after startup	Wait for about 300 frames of temporal data; it is expected that a new target has no label for a short time.
There are detection boxes but no skeleton	Check the BodyPose3DNet, the person crop and the engine loading; confirm that the engine exists under the `bodypose3dnet_vdeployable_accuracy_onnx_1.0` directory.
There is a skeleton but the label is unstable	Confirm the continuous input, the frame rate and the accumulation duration, and avoid looking at a single frame only.
The model label shows `sitting_down` / `standing` / `walking` and so on	These are the 6 classes of the model (sitting_down, getting_up, sitting, standing, walking, jumping); it is not an error.
The `PREPROC` / `POSE-CLASS` log	This is the normal informational output of the pose post-processing and the action classification, not an error.
No picture with USB	Check the node, the format, the occupancy, the permissions and the graphical session; the resolution has to match the capsfilter in the YAML.
No picture with CSI	The script directly uses the Argus; check the compatible module, the sensor driver, `nvargus-daemon`, the `sensor-id` and whether the camera is occupied.

Symptom	Check and handling
`No cameras available`	Check the CSI ribbon cable, the `sensor-id=0` and `nvargus-daemon`.

7. References

- NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html